

ALGORITHM
Autumn 2024, Quiz 9, May 15
11:45-12:05 am

姓名：

學號：

系所：

1. Consider a modification of the rod-cutting problem in which, in addition to a price p_i for each rod, each cut incurs a fixed cost of c . The revenue associated with a solution is now the sum of the prices of the pieces minus the costs of making the cuts. Give a dynamic-programming algorithm to solve this modified problem.

- (a) Suppose that each cut incurs a fixed cost of 2. Determine an optimal solution to cut a rod of length 10 with the following table of prices p_i for length $i = 1, 2, \dots, 10$.

length i	1	2	3	4	5	6	7	8	9	10
price p_i	1	3	6	6	10	9	8	13	8	12

- (b) Give a dynamic-programming algorithm to solve this modified problem.

Quiz9-(a)

- $p[i]$: length i 直接賣的價錢
- $r[i]$: length i 最好的價錢
- $s[i]$: 最左邊那一刀的位置

變數定義、初始化: 1分

length	0	1	2	3	4	5	6	7	8	9	10
$p[i]$	0	1	3	6	6	10	9	8	13	8	12
$r[i]$	0	1	3	6	6	10	10	11	14	14	18
$s[i]$	0	1	2	3	4	5	3	2	3	3	5

初始化 $r[0] = 0$ $s[0] = 0$

$$r[1] = \max\{p[1]\} = 1, s[1] = 1$$

子問題最佳解: 2分
問題最佳解: 2分

$$r[2] = \max\{p[1]+r[1]-2, p[2]\} = 3, s[2] = 2$$

$$r[3] = \max\{p[1]+r[2]-2, p[2]+r[1]-2, p[3]\} = 6, s[3] = 3$$

$$r[4] = \max\{p[1]+r[3]-2, p[2]+r[2]-2, p[3]+r[1]-2, p[4]\} = 6, s[4] = 4$$

$$r[5] = \max\{p[1]+r[4]-2, p[2]+r[3]-2, p[3]+r[2]-2, p[4]+r[1]-2, p[5]\} = 10, s[5] = 5$$

$$r[6] = \max\{p[1]+r[5]-2, p[2]+r[4]-2, p[3]+r[3]-2, p[4]+r[2]-2,$$

$$p[5]+r[1]-2, p[6]\} = 10, s[6] = 3$$

$$r[7] = \max\{p[1]+r[6]-2, p[2]+r[5]-2, p[3]+r[4]-2, p[4]+r[3]-2,$$

$$p[5]+r[2]-2, p[6]+r[1]-2, p[7]\} = 11, s[7] = 2$$

$$r[8] = \max\{p[1]+r[7]-2, p[2]+r[6]-2, p[3]+r[5]-2, p[4]+r[4]-2,$$

$$p[5]+r[3]-2, p[6]+r[2]-2, p[7]+r[1]-2, p[8]\} = 14, s[8] = 3$$

$$r[9] = \max\{p[1]+r[8]-2, p[2]+r[7]-2, p[3]+r[6]-2, p[4]+r[5]-2,$$

$$p[5]+r[4]-2, p[6]+r[3]-2, p[7]+r[2]-2, p[8]+r[1]-2, p[9]\} = 14, s[9] = 3$$

$$r[10] = \max\{p[1]+r[9]-2, p[2]+r[8]-2, p[3]+r[7]-2, p[4]+r[6]-2, p[5]+r[5]-2,$$

$$p[6]+r[4]-2, p[7]+r[3]-2, p[8]+r[2]-2, p[9]+r[1]-2, p[10]\} = 18, s[10] = 5$$

[解] 說明最佳解切割方式: 1分

當長度為10時, 查表 $s[10]=5$, 表示會在5等分切一刀, 此時右邊長度剩下5, 再查表 $s[5]=5$, 表示不用再切了, 所以在第5等分的位置切一刀, 會得到 $10 + 10 - 2 = 18$ 的最佳解。

Quize9-(b) 4分

- p: 直接賣的價錢
- n: rod 長度
- c: 每刀成本

```
1  cut_rod(p, n, c)
2      int r[n] // 最好的價錢
3      int s[n] // 最左邊那一刀
4      r[0] = 0
5      for j = 1 to n {
6          int q = INT_MIN
7          for i = 1 to j
8              int cur_price = p[i] + r[j - i]
9              if (j - i != 0) // 扣除每刀成本
10                 cur_price -= c
11                 if(q < cur_price) // 如果價錢更高
12                     q = cur_price // 更新價錢
13                     s[j] = i // 記住最左邊那一刀
14             r[j] = q; // 記住最好的價錢
15     return r and s;
```